



Comparative study of reference currents and DC bus voltage control for Three-Phase Four-Wire Four-Leg SAPF to compensate harmonics and reactive power with 3D SVM

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ABSTRACT

In this paper the performances of three reference currents and DC bus voltage control techniques for Three-Phase Four-Wire Four-Leg SAPF are compared for balanced and unbalanced load conditions. The main goals are to minimize the harmonics, reduce the magnitude of neutral current, eliminate the zero-sequence current components caused by single-phase nonlinear loads and compensate the reactive power, and on the other hand improve performances such as robustness, stabilization, trajectory pursuit, and reduce time response. The three techniques are analyzed mathematically and simulation results are compared. The techniques considered for comparative study are the PI Control, Sliding Mode Control and the Backstepping Control. Synchronous reference frame theory (SRF) in the dqo -axes is used to generate the reference currents, of the inverter.

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1. Introduction

Active power filters are a new solution for the compensation of harmonic pollution in power systems. Two types of active filters are distinguished: series and shunt. The shunt active power filter makes the compensation of harmonic currents and reactive power [1,2]. Analysis of the operation of a three-leg active filter connected to a four-wire network shows that it cannot compensate for perturbations due to single-phase nonlinear loads connected to a four-wire network. Harmonic currents circulating in the network remain partly compensated and amplitude of the zero sequence current is not reduced; to rectify it, a four-leg shunt active filter will be necessary [3–6].

Among the various topologies, the Three-Phase Four-Wire Four-Leg SAPF based on voltage source inverter is the most common one because of its good efficiency. Its performance depends on adaptive control approaches. There are four major parts of an active power filter controller. The first is the reference signal generation technique, the second is that switching signals generation techniques, the third is the controller of the system, and the fourth is the inverter to inject the compensating current into AC mains [4,7].

This paper presents PI, sliding mode and the Backstepping in the dqo -axes for the reference currents and the DC bus voltage control of a Three-Phase Four-Wire Four-Leg SAPF. The 3D-SVM technique will be used for generating switching signals. This technique not only inherits the fixation of the switching frequency of the inverter switches, but also provides a good dynamic response in the active filtering.

The PI controller used requires precise linear mathematical models, which are difficult to obtain, and fails to perform satisfactorily under parameter variations, nonlinearity, load disturbance, etc. It will cause DC voltage overshoot and inrush reference currents which will lead to protection or even equipment damage when SAPF is plunged into the system. The voltage overshoot and inrush current have been the constraints that restrict the development of an active power filter. The sliding mode control has been widely applied to power converters due to its operation characteristics such as fastness, robustness and stability for large load variations [5,8,9]. Indeed, this control technique rapidly rejects the variation of nonlinear loads and it can be concluded that the pursuit performances are little affected and there is remarkable robustness [10,15]. However, the sliding mode control using the system, it can excite the dynamics are not considered, such as the phenomenon of chattering in the direct current i_{dc} to the output of the DC voltage regulator and the reference voltages to the input of SVM-3D block, which is generated on the sliding surface by the function sign (the discontinuous control) [11], and generally undesirable

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because it added to the spectrum of the control of high frequency components, this phenomenon can excites not considered effects. Although this filtering phenomenon of sliding mode control reduces the excitement, it has the disadvantage of reducing at the same time the robustness of this technique [11,12]. Backstepping control technique discussed in this article considers the question of robustness, the response time is subdued in the control effort and results in the application of the Lyapunov function [8,9].

A short description of each control part of the SAPF with necessary equations is provided. In Section II, the main circuit principle of a Three-Phase Four-Wire Four-Leg SAPF is described. In Section III, the SRF algorithm of generating reference signals and zero sequence current is presented. In Section IV, description of each technique with necessary equations for the reference currents and the DC bus voltage control is briefly presented. In Section V, the simulation results are presented and comparative analysis is made based on the simulation results. The simulation is effected using SimPower Systems and S-Function of MATLAB/SIMULINK.

2. System modeling

Schematic block diagram of three-phase four-wire four-leg shunt active power filter with SRF and 3D-SVM control technique is shown in Fig. 1; the differential equations describing the dynamic model of the four-leg SAPF are defined in dqo -axes, as given in Eq. (1)

$$\begin{cases} \frac{di_{fd}}{dt} = -\frac{R_f}{L_f}i_{fd} + \omega i_{fq} + \frac{1}{L_f}v_{fd} - \frac{1}{L_f}v_{ld} \\ \frac{di_{fq}}{dt} = -\frac{R_f}{L_f}i_{fq} - \omega i_{fd} + \frac{1}{L_f}v_{fq} - \frac{1}{L_f}v_{lq} \\ \frac{di_{fo}}{dt} = -\frac{R_f}{L_f}i_{fo} + \frac{1}{L_f}v_{fo} - \frac{1}{L_f}v_{lo} \\ \frac{dV_{dc}}{dt} = -\frac{1}{C}i_{dc}^* \end{cases} \quad (1)$$

2.1. Synchronous reference frame theory (SRF)

The SRF theory gave very good results for the compensation of three-phase three-wire electricity network. For three-phase four-wire electricity networks, a theory based on the modification of the cross-vector theory of dqo -axes has been proposed [13]. Fig. 2 shows the basic principle of this SRF theory applied to a Three-Phase Four-Wire Four-Leg SAPF [14].

The SRF theory first requires the transformation of the three-phase load currents in $\alpha\beta o$ -axes as follows:

$$\begin{bmatrix} i_{l\alpha} \\ i_{l\beta} \\ i_{lo} \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} i_{l1} \\ i_{l2} \\ i_{l3} \end{bmatrix} \quad (2)$$

Currents in dqo -axes are obtained as

$$\begin{bmatrix} i_{ld} \\ i_{lqo} \\ i_{lq\alpha} \\ i_{lq\beta} \end{bmatrix} = \begin{bmatrix} 0 & \sin(\theta) & -\cos(\theta) \\ 0 & \cos(\theta) & \sin(\theta) \\ -\cos(\theta) & 0 & 0 \\ -\sin(\theta) & 0 & 0 \end{bmatrix}^{-1} \begin{bmatrix} i_{lo} \\ i_{l\alpha} \\ i_{l\beta} \end{bmatrix} \quad (3)$$

i_{ld} and i_{lq} currents can be expressed as the sum of two components, continuous and another alternative, such as

$$\begin{bmatrix} i_{ld} \\ i_{lq} \end{bmatrix} = \begin{bmatrix} \tilde{i}_{ld} \\ \tilde{i}_{lq} \end{bmatrix} + \begin{bmatrix} \bar{i}_{ld} \\ \bar{i}_{lq} \end{bmatrix} \quad (4)$$

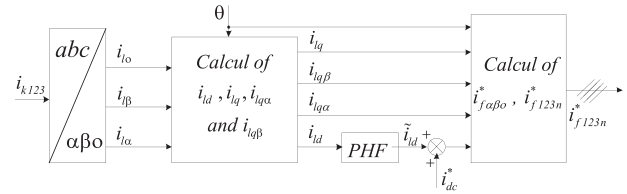


Fig. 2. Schematic diagram of the synchronous reference frame theory.

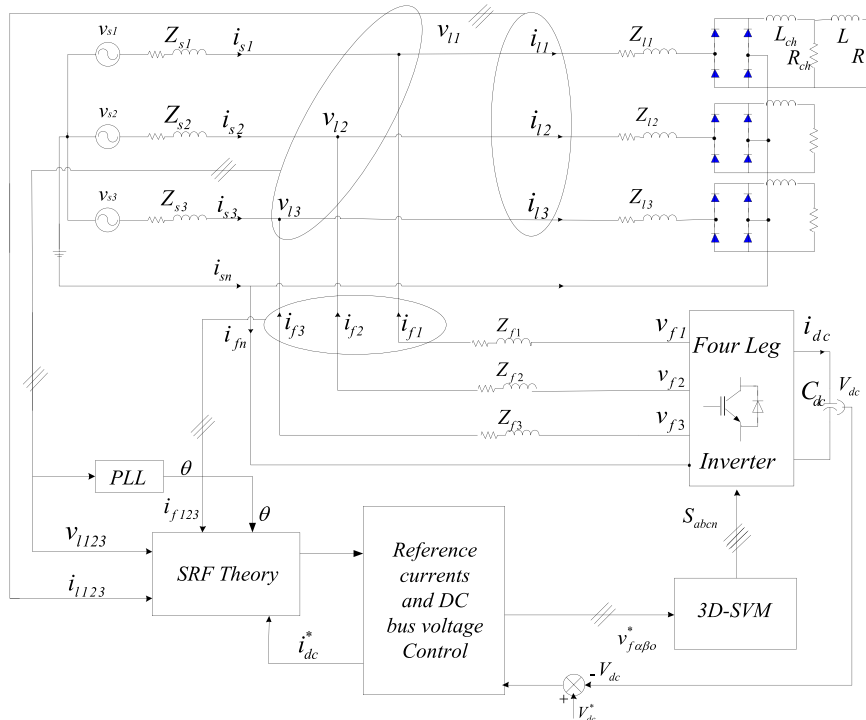


Fig. 1. Schematic block diagram of three-phase four-wire four-leg shunt active power filter with SRF and 3D-SVM control technique.

where $\bar{i}_{ld}$, $\bar{i}_{lq}$ are the direct components of i_{ld} and i_{lq} , and $\tilde{i}_{ld}$, $\tilde{i}_{lq}$ are the alternative components of i_{ld} and i_{lq} , respectively.

The filter reference currents are expressed in dqo -axes by

$$\begin{bmatrix} i_{ld}^* \\ i_{lq}^* \\ i_{lo}^* \end{bmatrix} = \begin{bmatrix} 0 & 0 & -\cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) & 0 & 0 \\ -\cos(\theta) & \sin(\theta) & 0 & 0 \end{bmatrix} \begin{bmatrix} i_{fd}^* \\ i_{fq}^* \\ i_{fq\alpha}^* \\ i_{fq\beta}^* \end{bmatrix} \quad (5)$$

with

$$\begin{bmatrix} i_{fd}^* \\ i_{fq}^* \\ i_{fq\alpha}^* \\ i_{fq\beta}^* \end{bmatrix} = \begin{bmatrix} i_{ld} \\ i_{lq} \\ i_{lq\alpha} \\ i_{lq\beta} \end{bmatrix}$$

The inverse Concordia transform calculates the reference currents in the abc -axes as follows:

$$\begin{bmatrix} i_{f1}^* \\ i_{f2}^* \\ i_{f3}^* \end{bmatrix} = \sqrt{\frac{2}{3}} \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & -\frac{\sqrt{3}}{2} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} i_{f\alpha}^* \\ i_{f\beta}^* \\ i_{fo}^* \end{bmatrix} \quad (6)$$

3. Technique for generating switching signals

The goal of the control is to generate the switching commands, so that current is injected by the inverter currents closest to the reference.

3.1. Three-dimensional space vector modulation with null vector

The major advantage of a four-leg inverter is that the DC bus utilization can be improved using 3D space vector modulation [16].

3.2. Prism identification

Six prisms in 3D space can be identified and numbered as Prisms I through VI (Fig. 3). Within the selected prism, there are six non-zero switching state vectors and two zero switching state vectors. Fig. 4 shows the physical positions of the switching state vectors in $\alpha\beta o$ -axes [17–19].

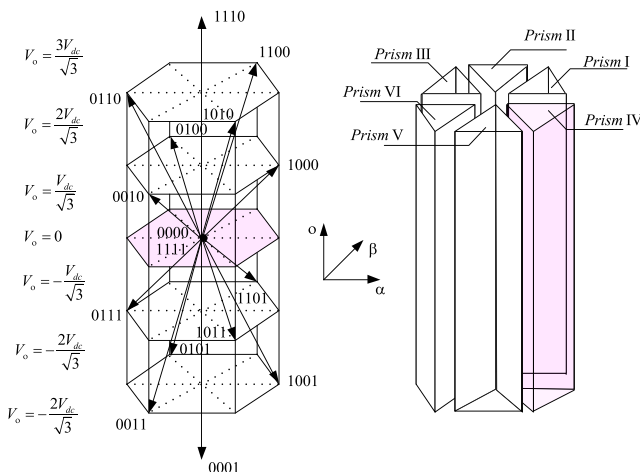


Fig. 3. Voltage vectors in the $\alpha\beta o$ -axes, and the selection of prism.

3.3. Tetrahedron identification

The tetrahedron is formed by three non-zero voltage vectors and two other zero vectors, as shown in Fig. 4. Fig. 5 describes the representation of tetrahedron 1 plans belonging to prism I [17,18].

3.4. Duty cycle calculation

For the tetrahedron i of the prism j , the states vectors of the inverter ($v_x, v_y, v_z, v_1, v_{16}$) are the adjacent vectors to the reference voltage vector. These vectors are applied individually for certain periods t_x, t_y, t_z and t_0 , such that the vector $\bar{v}^*$ is equal to the mean value of those vectors for a period of switching [20]

$$\bar{v}^* = \bar{v}_x + \bar{v}_y + \bar{v}_z + \bar{v}_0 \quad (7)$$

The mean reference vector is expressed by

$$\bar{v}^* = \frac{1}{T_h} \int_t^{t+T_h} \bar{v}^* dt \quad (8)$$

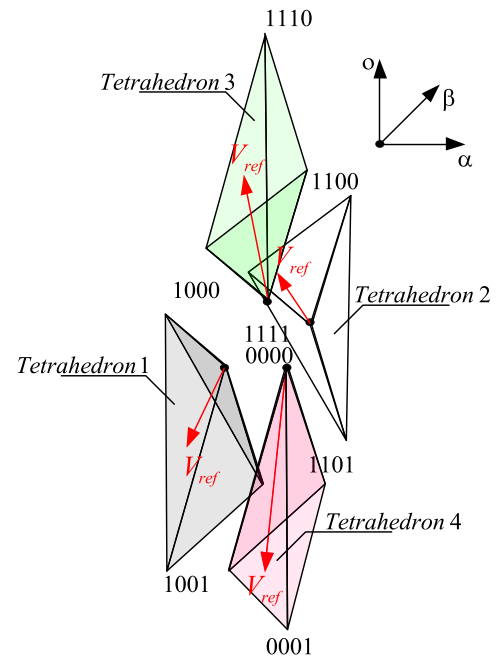


Fig. 4. Representation of tetrahedra in prism I.

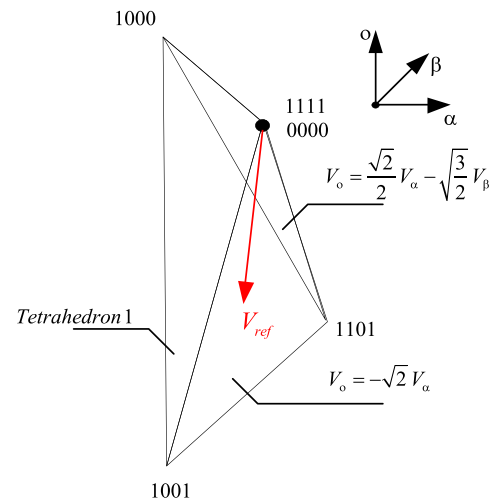


Fig. 5. Projection of the reference vector on the adjacent vectors.

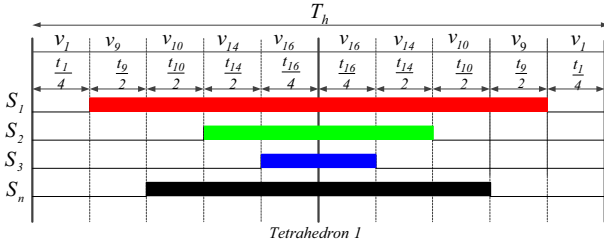


Fig. 6. Principle of generation in the pulses by 3D-SVM.

As the period of switching is very small, the mean voltage value can be considered as constant; and as the vectors V_x , V_y , V_z , and V_o are fixed, it follows that

$$\bar{V}_i = \frac{1}{T_h} \int_{t_i}^{t_i+T_h} V_i dt = \frac{1}{T_h} V_i t_i \quad (9)$$

where $i=x, y, z, o$.

Then

$$T_h V^* = t_x V_x + t_y V_y + t_z V_z + t_o V_o \quad (10)$$

$$t_o = T_h - t_x - t_y - t_z \quad (11)$$

3.5. The control pulses generation

Fig. 6 shows, on a switching period, the distribution of voltage vectors to be applied in the first prism tetrahedron [16] (Figs. 4 and 5).

4. Controllers synthesis

4.1. Sliding mode control

The system established in (1) is a multi-input multi-output nonlinear system, to apply the sliding mode control on the three-phase four-wire four-leg shunt active filter; system (1) should be divided into two subsystems as follows [21]:

• Subsystem (A)

In the first subsystem defined by Eq. (12), the voltages v_{fd}^* , v_{fq}^* and v_{fo}^* are selected as control variables, and the currents i_{fd} , i_{fq} and i_{fo} as the outputs filter [8]

$$\begin{cases} \frac{di_{fd}}{dt} = -\frac{R_f}{L_f} i_{fd} + \omega i_{fq} + \frac{1}{L_f} v_{fd}^* - \frac{1}{L_f} v_{ld} \\ \frac{di_{fq}}{dt} = -\frac{R_f}{L_f} i_{fq} - \omega i_{fd} + \frac{1}{L_f} v_{fq}^* - \frac{1}{L_f} v_{lq} \\ \frac{di_{fo}}{dt} = -\frac{R_f}{L_f} i_{fo} + \frac{1}{L_f} v_{fo}^* - \frac{1}{L_f} v_{lo} \end{cases} \quad (12)$$

• Subsystem (B)

The equation that describes this subsystem is defined by

$$\frac{dV_{dc}}{dt} = \frac{1}{C_{dc}} i_{dc}^* \quad (13)$$

In this subsystem, the current i_{dc}^* is considered as control variable and the voltage V_{dc} as an output variable.

4.1.1. Synthesis controllers

4.1.1.1. Control currents. From the system of Eq. (12), three surfaces are defined

$$\begin{cases} S(i_{fd}) = k_d(i_{fd} - i_{fd}^*) + k_{id} \int (i_{fd} - i_{fd}^*) dt \\ S(i_{fq}) = k_q(i_{fq} - i_{fq}^*) + k_{iq} \int (i_{fq} - i_{fq}^*) dt \\ S(i_{fo}) = k_o(i_{fo} - i_{fo}^*) + k_{io} \int (i_{fo} - i_{fo}^*) dt \end{cases} \quad (14)$$

A) Equivalent control

During the sliding mode

$$\begin{cases} S(i_{fd}) = S(i_{fq}) = S(i_{fo}) = 0 \\ \dot{S}(i_{fd}) = \dot{S}(i_{fq}) = \dot{S}(i_{fo}) = 0 \end{cases} \quad (15)$$

From [20]

$$\dot{X} = AX + Bu + D \quad (16)$$

and

$$S(X) = K(X^* - X) + K_i \int (X^* - X) dt \quad (17)$$

with

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} i_{fd} \\ i_{fq} \\ i_{fo} \end{bmatrix}; \quad A = \begin{bmatrix} -\frac{R_f}{L_f} & \omega & 0 \\ \omega & -\frac{R_f}{L_f} & 0 \\ 0 & 0 & -\frac{R_f}{L_f} \end{bmatrix}; \quad B = \frac{1}{L_f}; \quad D = \begin{bmatrix} -\frac{1}{L_f} v_{ld} \\ -\frac{1}{L_f} v_{lq} \\ -\frac{1}{L_f} v_{lo} \end{bmatrix}$$

Solving Eq. (23) gives the expression of the equivalent command as follows:

$$u_{eq(dqo)} = (K_{dqo} B)^{-1} [K_{idqo} (X_{fdqo}^* - X_{fdqo}) + K_{dqo} (\dot{X}_{fdqo}^* - (AX_{fdqo} + D))] \quad (18)$$

$$k = \begin{bmatrix} k_d & 0 & 0 \\ 0 & k_q & 0 \\ 0 & 0 & k_o \end{bmatrix}; \quad k_i = \begin{bmatrix} k_{id} & 0 & 0 \\ 0 & k_{iq} & 0 \\ 0 & 0 & k_{io} \end{bmatrix}$$

B) Discontinuous control

If the trajectories are different from the state of the sliding surface, the discontinuous control ensures decrease of distance between the state trajectory and the sliding surface. This command is selected so as to ensure the convergence of the path to its reference. In actual case, the discontinuous control function is selected as [10,8]

$$u_{dis(dqo)} = \begin{bmatrix} u_{dis(d)} \\ u_{dis(q)} \\ u_{dis(o)} \end{bmatrix} = \text{sign}(S(X)) U_{\max} \quad (19)$$

$$u_{dis(dqo)} = \begin{cases} U_{\max} & \text{if } S(x_{1,2}) > 0 \\ -U_{\max} & \text{if } S(x_{1,2}) < 0 \\ 0 & \text{if } S(x_{1,2}) = 0 \end{cases} \quad (20)$$

The control by sliding mode is given by equation 21 and Fig. 7

$$u = u_{eq(dqo)} + u_{dis(dqo)} = \begin{bmatrix} v_{fd}^* \\ v_{fq}^* \\ v_{fo}^* \end{bmatrix} \quad (21)$$

with

$$k_{id} = k_{iq} = k_{io} = 8 \times 10^5, \quad k_d = k_q = k_o = 10^{-8},$$

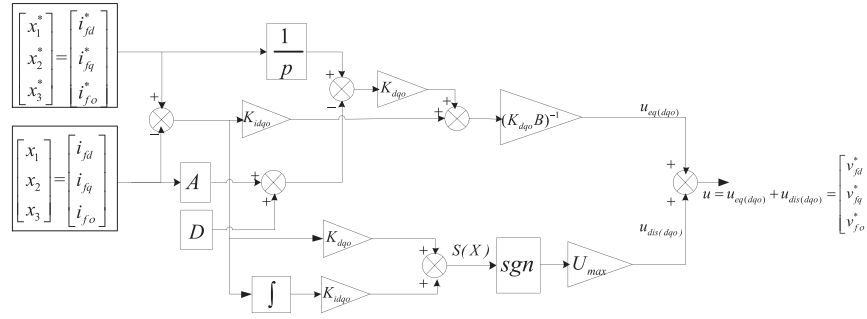


Fig. 7. Schematic diagram of the currents controlled by the sliding mode technical.

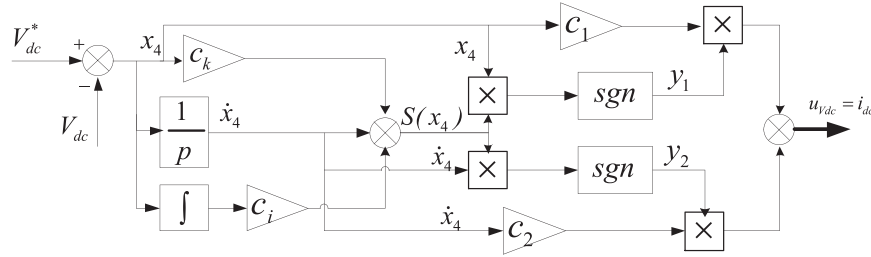


Fig. 8. Schematic diagram of the DC bus voltage controlled by the sliding mode technical.

$$U_{max\alpha} = U_{max\beta} = U_{max0} = 1e^3$$

4.1.1.2. DC bus voltage regulation. The fourth surface is that of the DC bus voltage, where the sliding surface defined by Golwala and Chudamani [20] and Liu et al. [25] is chosen

$$S(x_4) = c_k x_4 + \dot{x}_4 + c_i \int x_4 \quad (22)$$

with

$$x_4 = \varepsilon(V_{dc}) \quad (23)$$

In the sliding mode, switching functions are defined as

$$y_1 = \begin{cases} 1 & \text{if } S(x_4)x_4 > 0 \\ -1 & \text{if } S(x_4)x_4 < 0 \end{cases} \quad (24)$$

$$y_2 = \begin{cases} 1 & \text{if } S(x_4)\dot{x}_4 > 0 \\ -1 & \text{if } S(x_4)\dot{x}_4 < 0 \end{cases} \quad (25)$$

Hence the output of the sliding mode controller is defined by equation 24 and Fig. 8

$$u_{vdc} = i_{dc} = c_1 x_4 y_1 + c_2 \dot{x}_4 y_2 \quad (26)$$

with

$$c_k = c_i = 1e^3, \quad c_1 = 0.4, \quad c_2 = 5e^{-6}.$$

4.2. Backstepping control

Backstepping control is a systematic and flexible technique, newly developed and can be effectively used to linearize a nonlinear system in the presence of uncertainties [8], and the control laws are derived easily step by step across an appropriate Lyapunov function. Contrary to the sliding mode control and other nonlinear techniques such as feedback, the input output linearization which is based on the exact cancellation of nonlinearity in the system [22–28].

The Backstepping control takes into account the nonlinearity in the conception phase of the control and provides efficient process

in conception control for both the control and tracking problems of uncertain parameters [12].

In this work, virtual controls are known because the reference currents that stabilize the operation of the Three-Phase Four-Wire Four-Leg SAPF are the currents i_{fd} , i_{fq} , i_{fo} . The purpose of control is to obtain a relevant constructively control law that ensures the stability of the closed loop system via a Lyapunov function. To validate this technique, the reference currents and DC bus voltage controls of a Three-Phase Four-Wire Four-Leg SAPF is applied.

As the sliding mode control, to apply the Backstepping control on the Three-Phase Four-Wire Four-Leg SAPF, system (1) is divided into two subsystems, as discussed next.

4.2.1. Subsystem (A)

In the first subsystem defined by the two Eq. (27), and to find the direct component of the reference current i_{fd}^* of the three phase four wire four-leg SAPF, the direct current i_{dc} of the output regulator DC bus voltage will be subtracted from the direct harmonic current of the nonlinear loads; in this case, the reference voltage v_{fd}^* and the direct current i_{dc} are chosen as control variables, while the current i_{fd} and the DC bus voltage are the output variables.

$$\begin{cases} \frac{di_{fd}}{dt} = -\frac{R_f}{L_f} i_{fd} + \omega i_{fq} + \frac{1}{L_f} v_{fd}^* - \frac{1}{L_f} v_{ld} \\ \frac{dV_{dc}}{dt} = \frac{1}{C_{dc}} i_{dc} \end{cases} \quad (27)$$

Subsystem 1, corresponding to the two control variables and the two output variables, is given by the two state matrix forms as follows:

$$\begin{bmatrix} \dot{x}_0 \\ \dot{x}_4 \\ \dot{x}_1 \end{bmatrix} = \begin{bmatrix} V_{dc} \\ \varepsilon(V_{dc}) \\ i_{fd} \end{bmatrix} \quad (28)$$

$$\begin{bmatrix} u_{dc} \\ u_d \end{bmatrix} = \begin{bmatrix} i_{dc} \\ v_{fd}^* \end{bmatrix} \quad (29)$$

4.2.2. Subsystem (B)

To compensate the reactive power, reducing the harmonics and zero sequence, the two components of load current i_{lq} and zero sequence current i_{lo} are used as reference current i_{fq}^*, i_{fo}^* , respectively, of the SAPF, and considered as output variables. Also for the control variables, the two reference voltages v_{fq}^*, v_{fo}^* are used. The equations describing this subsystem are defined by

$$\begin{cases} \frac{di_{fq}}{dt} = -\frac{R_f}{L_f}i_{fq} - \omega i_{fd} + \frac{1}{L_f}v_{fq}^* - \frac{1}{L_f}v_{lq} \\ \frac{di_{fo}}{dt} = -\frac{R_f}{L_f}i_{fo} + \frac{1}{L_f}v_{fo}^* - \frac{1}{L_f}v_{lo} \end{cases} \quad (30)$$

Subsystem B, which defines the two control variables v_{fq}^*, v_{fo}^* , and the two output variables i_{fq}, i_{fo} , is given by the two following matrix forms [24]:

$$\begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} i_{fq} \\ i_{fo} \end{bmatrix} \quad (31)$$

$$\begin{bmatrix} u_q \\ u_o \end{bmatrix} = \begin{bmatrix} v_{fq}^* \\ v_{fo}^* \end{bmatrix} \quad (32)$$

4.2.3. Regulation of the current i_{fd} and DC bus voltage V_{dc}

For the first subsystem, the Backstepping errors e_1, e_2 are defined as

$$e_1 = V_{dc}^* - V_{dc} = x_4 = x_o^* - x_o \quad (33)$$

$$e_2 = i_{fd}^* - i_{fd} - k_{V_{dc}}e_1 = x_1^* - x_1 - k_{V_{dc}}e_1 \quad (34)$$

The derivative of these errors is given as

$$\dot{e}_1 = \dot{x}_o^* - \dot{x}_o = \dot{V}_{dc}^* - \frac{1}{C_{dc}}i_{dc} \quad (35)$$

$$\dot{e}_2 = \dot{x}_1^* - \dot{x}_1 - k_{V_{dc}}\dot{e}_1 \quad (36)$$

where $k_{V_{dc}}$ is a positive constant.

In order to find the control variable u_d (i_{dc}), Eq. (34) is considered and the Lyapunov function is introduced

$$V_1(e_2) = \frac{1}{2}e_2^2 \quad (37)$$

The time derivative of V_1 is given by

$$\dot{V}_1(e_2) = e_2\dot{e}_2 \quad (38)$$

From Eqs. (35) and (36)

$$\dot{V}_1(e_2) = e_2(\dot{x}_1^* - \dot{x}_1 - k_{V_{dc}}(\dot{x}_o^* - \dot{x}_o)) \quad (39)$$

Also to ensure the stability of the first subsystem (the derivative of V_1 is negative), and the error converges to zero

$$\dot{e}_2 = -k_1 e_2. \quad (40)$$

where k_1 is a positive constant.

From the above equation and Eq. (34)

$$\dot{x}_1^* - \dot{x}_1 - k_{V_{dc}}\dot{x}_4 = -k_1(x_1^* - x_1 - k_{V_{dc}}x_4) \quad (41)$$

Because of this, the control variable u_{dc} is

$$u_{dc} = i_{dc} = k_{V_{dc}}(x_o^* - x_o) = k_{V_{dc}}(V_{dc}^* - V_{dc}) \quad (42)$$

And from Eqs. (27) and (41), the control variable u_{fd} is written as

$$u_{fd} = L_f k_1 x_1^* - L_f k_1 x_1 - L_f k_1 k_{V_{dc}} x_4 + L_f \dot{x}_1^* + R_f x_1 - L_f \omega x_2 - L_f k_{V_{dc}} \dot{x}_4 + v_{ld} \quad (43)$$

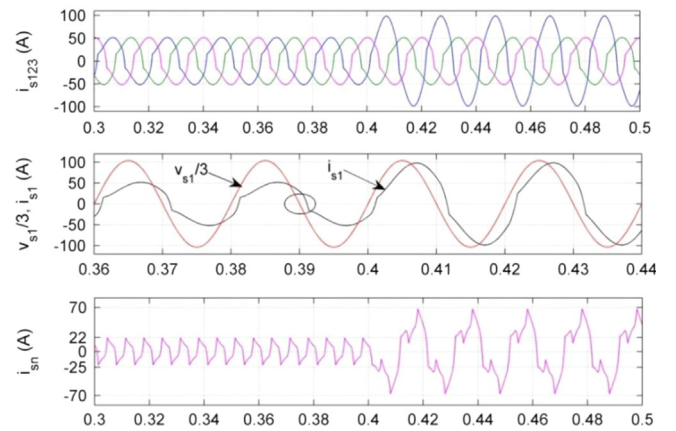


Fig. 10. Simulation results before filtering.

Table 1

The conditions considered for simulation.

PI	Sliding mode	Backstepping
$f_s = 14$ kHz	$f_s = 14$ kHz	$f_s = 14$ kHz
$V_{dcref} = 800$ V	$V_{dcref} = 800$ V	$V_{dcref} = 800$ V
$f_c = 10$ kHz	$k_{id} = k_{iq} = k_{io} = 5e^5$	$K_1 = k_2 = k_3 = 8e^6$
	$k_d = k_q = k_o = 2e^{-8}$	
	$u_{dmax} = u_{qmax} = u_{omax} = 1e^3$	
	$c_k = c_i = 1e^3$	
	$c_1 = 0.4, c_2 = 5e^{-6}$	
		$K_{V_{dc}} = 100$

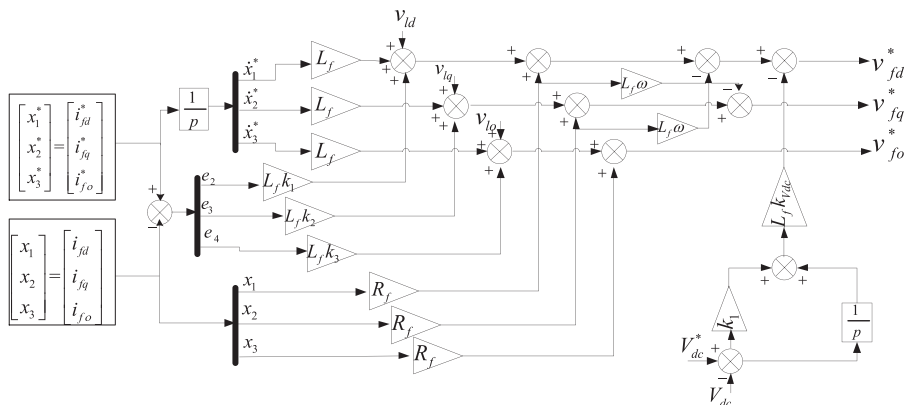


Fig. 9. Schematic diagram of the currents and DC bus voltage controlled by the Backstepping technical.

By replacing u_d by v_{fd}^* in Eq. (43), finally we get

$$u_{fd} = v_{fd}^* = L_f k_1 e_2 + L_f \ddot{x}_1^* + R_f x_1 - L_f \omega x_2 - L_f k_1 k_{V_{dc}} e_1 - L_f k_{V_{dc}} \dot{e}_1 + v_{ld} \quad (44)$$

4.2.4. Regulation of currents i_{fd} , i_{fo}

In the second subsystem as the previous, two reference voltages v_{fq}^* , v_{fo}^* are used as control variables, and the two current references of the SAPF i_{fq} , i_{fo} as output variables; the Backstepping

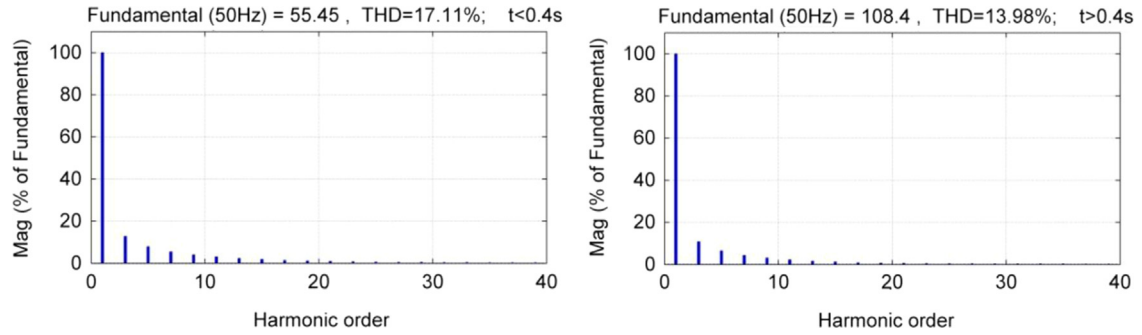


Fig. 11. Magnitude spectrum of source currents before filtering: before unbalanced loads; after unbalanced loads.

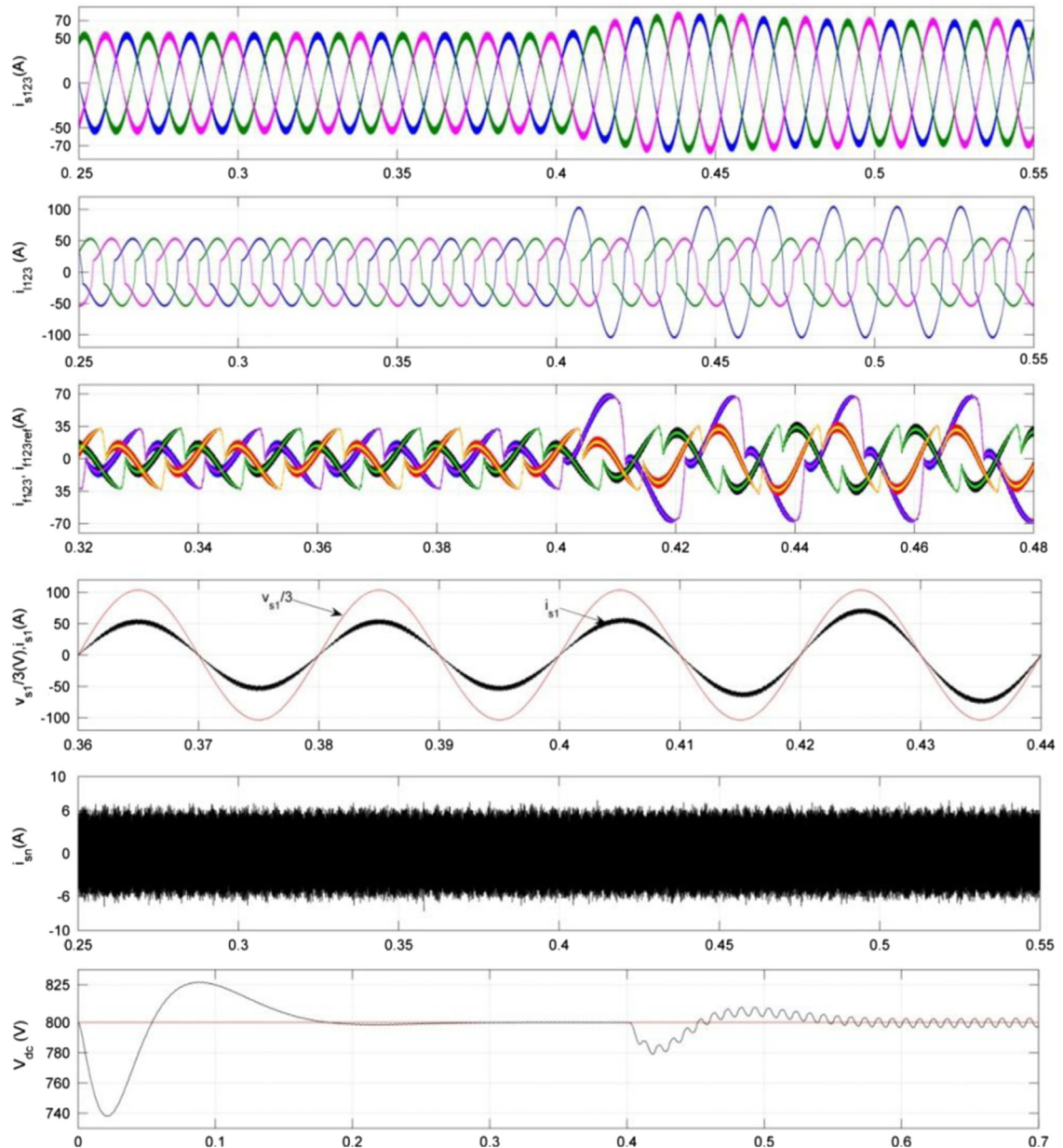


Fig. 12. Performance of the Three-Phase Four-Wire Four-Leg SAPF before and after unbalanced loads, using PI control.

errors e_3, e_4 in this step are given by the two following equations:

$$e_3 = i_{fq}^* - i_{fq} = x_2^* - x_2 \quad (45)$$

$$e_4 = i_{fo}^* - i_{fo} = x_3^* - x_3$$

The derivative of the error e_3 is given as

$$\dot{e}_3 = \dot{x}_2^* - \dot{x}_2 \quad (46)$$

To find the control variable u_q , the Lyapunov function is as follows:

$$V_2 = \frac{1}{2} e_3^2 \quad (47)$$

The time derivative of V_2 is given by

$$\dot{V}_2 = e_3 \dot{e}_3 \quad (48)$$

From (46) and (48)

$$\dot{V}_2(e_3) = e_3(\dot{x}_2^* - \dot{x}_2) \quad (49)$$

To ensure the stability of the first subsystem (the derivative of V_2 is negative), and the error converges to zero

$$\dot{e}_3 = -k_2 e_3 \quad (50)$$

where k_2 is a positive constant.

From the previous Eq. (46)

$$\dot{x}_2^* - \dot{x}_2 = -k_2(x_2^* - x_2) \quad (51)$$

As a result, and from (51), the control variable u_q is written as

$$u_q = L_f \dot{x}_2^* + R_f x_2 + L_f \omega x_1 + L_f k_2 e_3 + v_{lq} \quad (52)$$

Replacing u_q with v_{fq}^* in Eq. (52) finally gives

$$u_q = v_{fq}^* = L_f \dot{x}_2^* + R_f x_2 + L_f \omega x_1 + L_f k_2 e_3 + v_{lq} \quad (53)$$

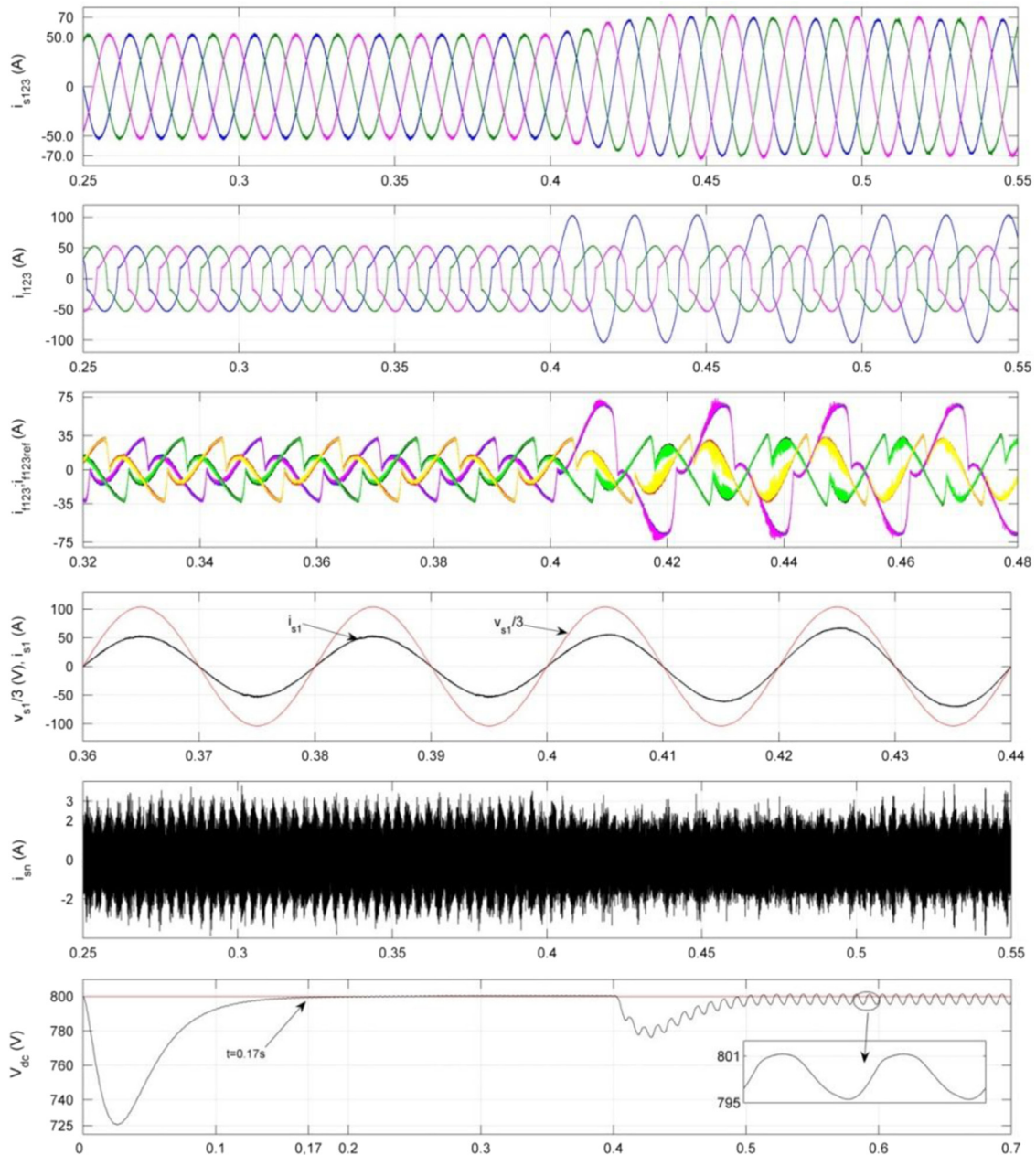


Fig. 13. Performance of the Three-Phase Four-Wire Four-Leg SAPF before and after unbalanced loads, using sliding mode control.

And as the two control variables u_d and u_q . The corresponding variable control u_o of the output i_{fo} (zero sequence) is given by

$$u_o = v_{fo}^* = L_f \dot{x}_3^* + R_f x_3 + L_f k_3 e_4 + v_{lo} \quad (54)$$

where

$$e_4 = i_{fo}^* - i_{fo} = x_3^* - x_3 \quad (55)$$

k_3 is a positive constant

The resulting control laws $u_{dqo} = v_{dqo}^*$ are designed to negatively define the derivative of Lyapunov functions, thus ensuring the convergence of the desired reference $x_i \simeq x_i^*$, and finally giving

$$u_{dqo} = \begin{cases} v_{fd}^* = L_f k_1 e_2 + L_f \dot{x}_1^* + R_f x_1 - L_f \omega x_2 - L_f k_1 k_{v_{dc}} e_1 - L_f k_{v_{dc}} \dot{e}_1 + v_{ld} \\ v_{fq}^* = L_f \dot{x}_2^* + R_f x_2 + L_f \omega x_1 + L_f k_2 e_3 + v_{lq} \\ v_{fo}^* = L_f \dot{x}_3^* + R_f x_3 + L_f k_3 e_4 + v_{lo} \end{cases} \quad (56)$$

Fig. 9 shows the block diagram of the Backstepping technique used to reference currents and DC bus voltage controls for the Three-Phase Four-Wire Four-Leg SAPF.

5. Simulation model of four-wire three-level four-leg SAPF

The simulation model of a Three-Phase Four-Wire Four-Leg SAPF was built using the *SimPower Systems* and *S-Function of MATLAB* from the structure shown in Fig. 1 before and after.

5.1. Before filtering

Fig. 10 shows the current waveform of the loads. It is a highly non-sinusoidal current and is deformed and it is not in sync with the corresponding voltages (power factor is $F_p=0$); the form of the neutral current has a maximum value of 22 A in the balanced case and 70 A in the unbalanced case.

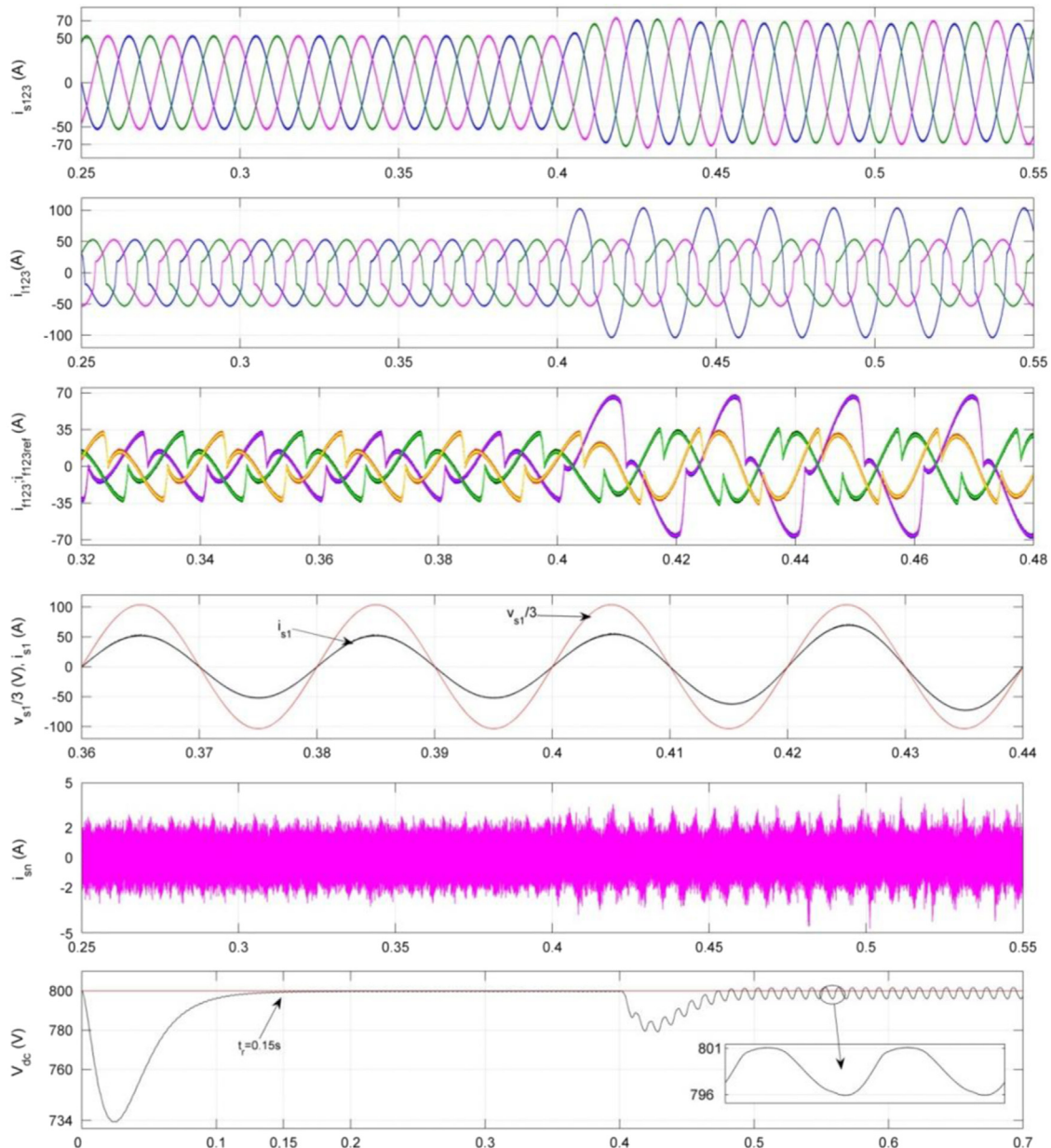


Fig. 14. Performance of the Three-Phase Four-Wire Four-Leg SAPF before and after unbalanced loads, using Backstepping control.

Fig. 11a and b shows the first phase source current's THD. Before unbalanced load ($t < 0.4$ s), the total harmonic distortion (THD) is 13.92%, when after ($t > 0.4$ s) it is 10.51%.

5.2. After filtering

To validate the proposed strategies in this paper, the results of numerical simulation are presented in this section. These results were obtained by applying the Three-Phase Four-Wire Four-Leg SAPF with the Three Dimensional Space Vector Modulation (3D-SVM) strategy, with the conditions considered for simulation specifications given in Table 1.

The simulation results of the system after compensation with the three reference currents and DC bus voltage control techniques before and after unbalanced loads are given in Figs. 12–17.

Figs. 12–14 illustrate the Three-Phase Four-Wire Four-Leg SAPF performance under different loads and controller conditions are fed by a three-dimensional space vector in order to have a constant switching of inverter switches and the minimizing switching losses of this inverter. The source current waveform after filtering is sinusoidal and in phase with the corresponding voltages (power factor is $F_p=1$); the monitoring of the current filter trajectory of i_{f123} and $i_{f123ref}$ is constructed with a zero error for the three techniques.

The amplitude of the neutral current is considerably reduced for ± 6 A in the PI controller to ± 3 A for the Sliding Mode and to ± 2 A

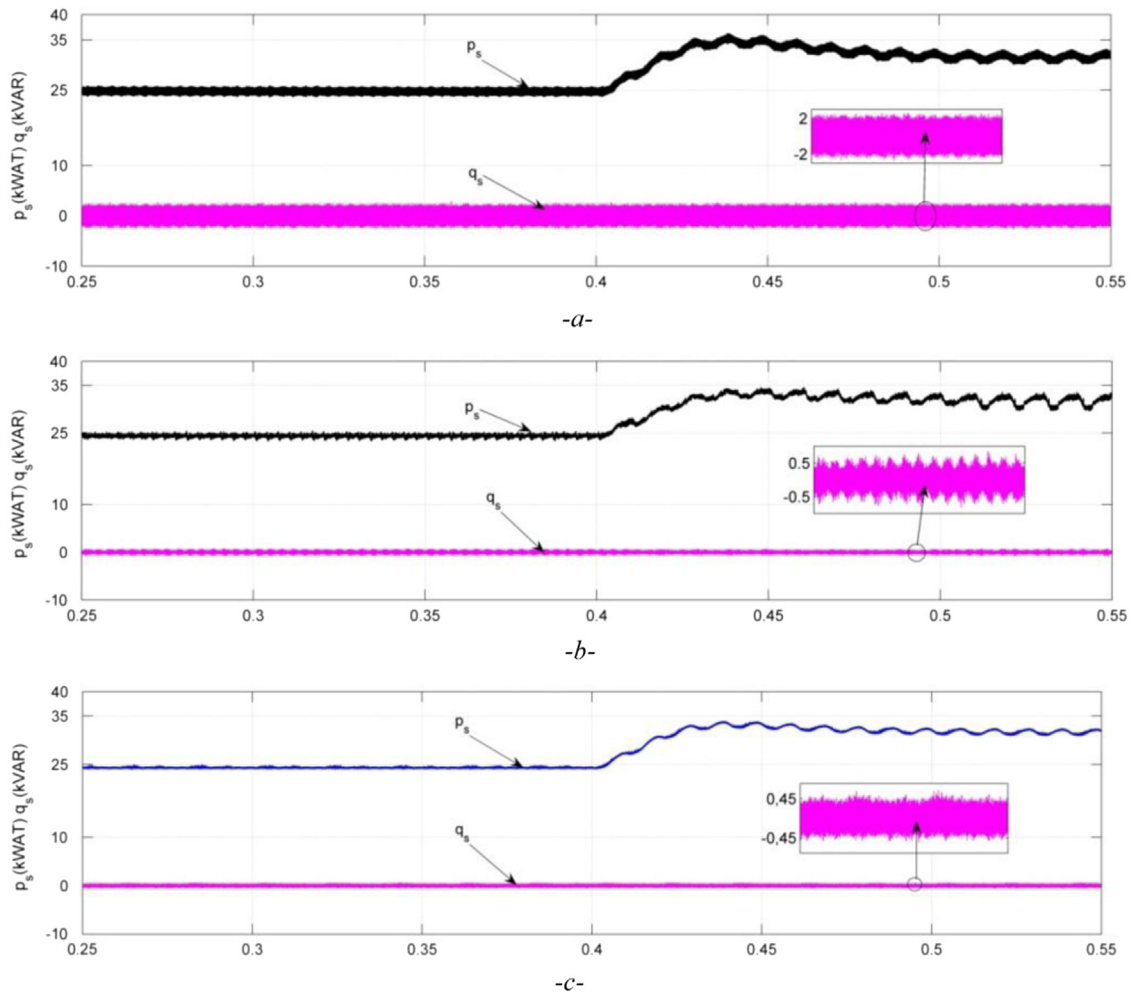


Fig. 15. The real and imaginary powers in the source before and after unbalanced loads. (a) PI control, (b) sliding mode control, and (c) Backstepping control.

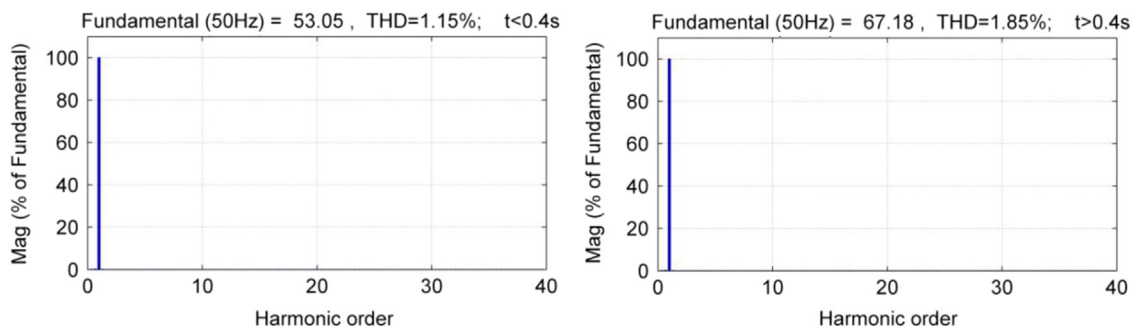


Fig. 16. THD of source currents using PI control: before and after unbalanced loads.

for the Backstepping controller, which is 10% of the neutral current before compensation, which also shows that the amplitude of the zero sequence (harmonics of order) is successfully reduced to 0.66 A for the Backstepping controller. The DC voltage precisely follows its reference with zero exceeding for the Sliding Mode and Backstepping controller, and the descent of this DC bus voltage at the boot up of the operating SAPF is 65 V with PI, 43 V with Sliding mode and 40 V with Backstepping control.

For the three control techniques, the real power in the source is constant with an increase after unbalance of nonlinear loads, and the imaginary power is almost zero (Fig. 15(a–c)), which means that the source has a power factor of almost unity before and after the unbalance of nonlinear loads, as well as smaller

undulations than in the control by Backstepping and in the range of ± 0.4 kVar.

Figs. 16–18 illustrate the harmonic spectrum of source current for the three techniques; before unbalanced load, THD is 1.15% with PI, 0.85% with Sliding mode and 0.73% with Backstepping control and after unbalanced load it is 1.85% with PI, 1.75% with Sliding mode and 1.68% with Backstepping control.

5.3. Robust Backstepping controller under four-leg SAPF parameters variations

In order to verify the robustness of the Backstepping controller under four-leg SAPF parameter variations, we carried out a test for a

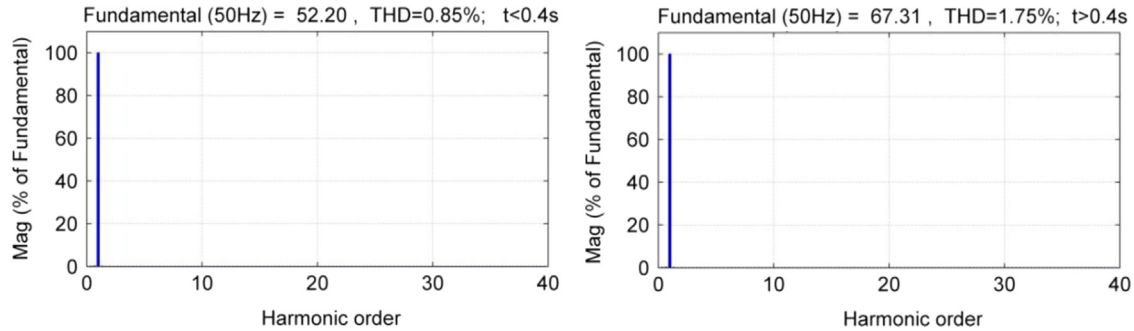


Fig. 17. THD of source currents using sliding mode control: before and after unbalanced loads.

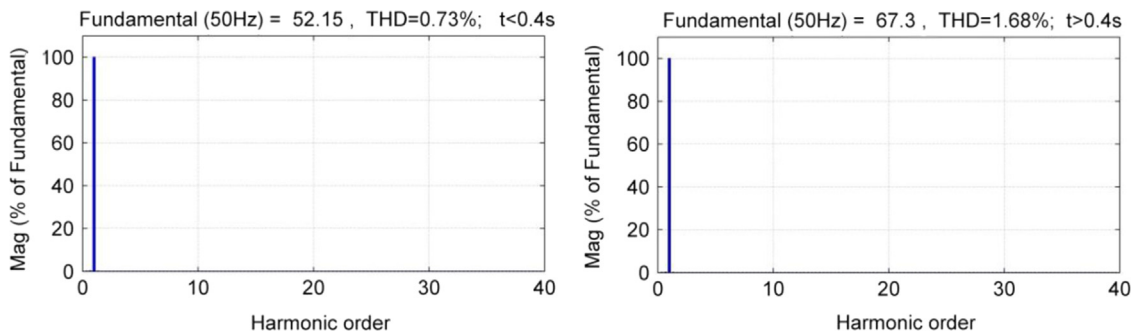


Fig. 18. THD of source currents using Backstepping control: before and after unbalanced loads.

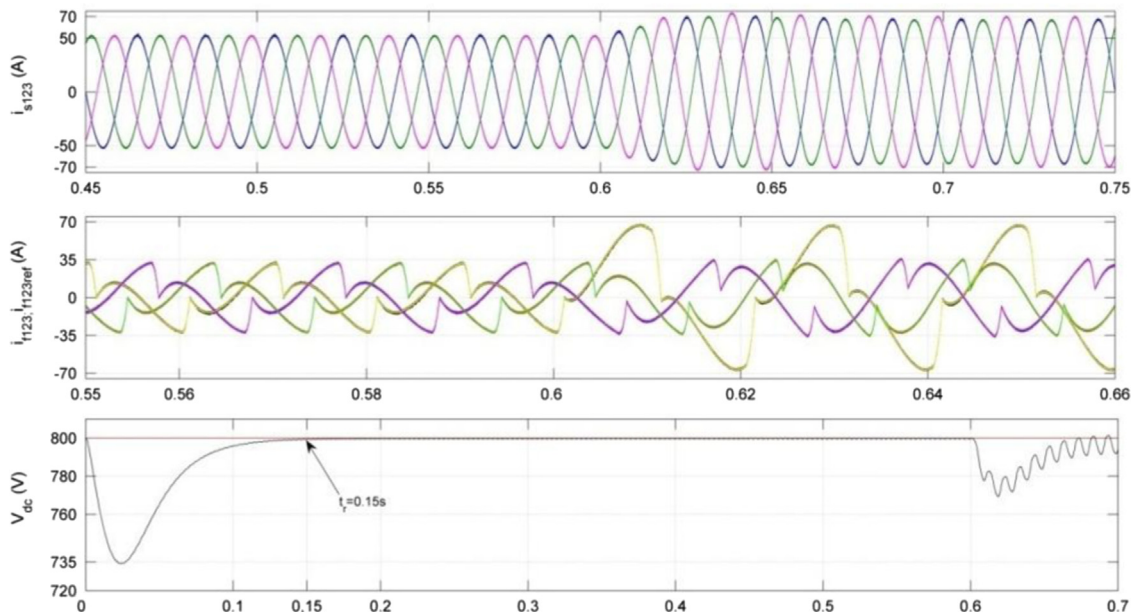


Fig. 19. Performance of the Three-Phase Four-Wire Four-Leg SAPF using Backstepping control before and after unbalanced loads, under four-leg SAPF parameters variations.

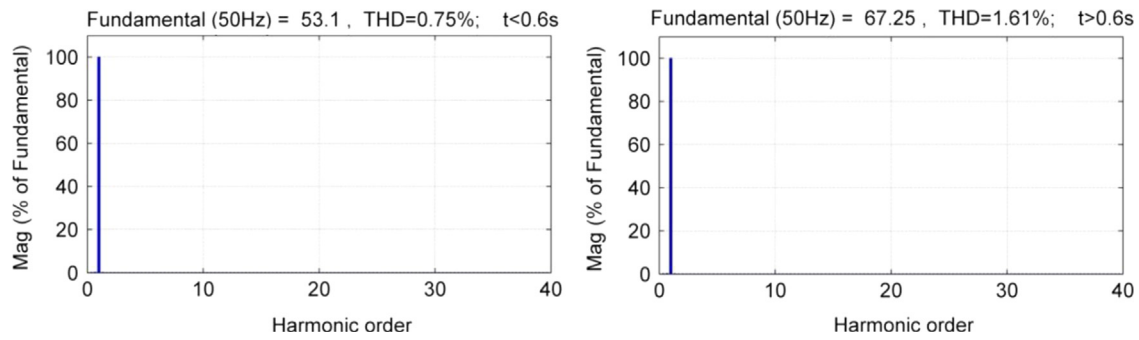


Fig. 20. THD of source currents using Backstepping control under four-leg SAPF parameters variations: before unbalanced loads, after unbalanced loads.

Table 2

Comparison of the different techniques.

	PI		Sliding mode		Backstepping	
	Balanced loads	Unbalanced loads	Balanced loads	Unbalanced loads	Balanced loads	Unbalanced loads
Source current THD (%)	1.15	1.85	0.85	1.75	0.73	1.68
The amplitude of the i_{sn} (A)	6		3		2	
Descent of the DC voltage (V)	65		75		65	
Exceeding	15V		Zero		Zero	
The response time (s)	0.11		0.17		0.15	
Undulations in the imaginary power (kVar)	± 2		± 0.5		± 0.4	

variation of 50% in the values of the capacitor C_{dc} and the impedance Z_f at time $t=2$ s. The unbalanced loads are applied at $t=6$ s. Fig. 19 shows in order the source currents, the filter currents, and the DC bus voltage. Fig. 20 illustrates the harmonic spectrum of source currents; before unbalanced load, THD is 0.75%, and after unbalanced load it is 1.61%. The results indicate that the controller has very good dynamics and a very good precision.

Table 2 gives the THD, amplitude of the neutral current, and exceeding and rippling of the DC voltage comparison of PI with sliding mode control strategy using Matlab/Simulink.

6. Conclusion

In this paper, the proposed Backstepping technique is compared with PI and sliding mode control. From the results obtained it is seen that the compensation performance of Three-Phase Four-Wire Four-Leg SAPF with three-dimensional space vector modulation (3D-SVM strategy in order to have a constant switching of inverter switches and the minimizing of switching losses of this inverter) is almost similar to all three techniques; however, the current flowing in the neutral line of the balanced and unbalanced nonlinear loads and the THDs are reduced in the proposed Backstepping technique, and by compensating harmonics and reactive power requirement the power quality has been improved in this technique.

The Backstepping technique control provides notable improvements, better dynamic performance and good robustness over conventional PI and sliding mode controllers. Indeed, the Backstepping technique offers good static and dynamic performance resulting in a shorter response without overshoot.

Finally, the simulation results validated both the steady state and dynamic behavior of the proposed Backstepping technique. The simulation results show that four-leg SAPF with a Backstepping technique presents very good performances from the point of view robustness. The four-leg SAPF with a Backstepping technique also offers a good dynamic and very good precision. The THD of the supply current with all algorithms is well below the limit imposed by the IEEE-519 or CEI 61000 standards.

Appendix A. Simulation Parameters

The system parameters considered for simulation and the load specifications are given in Table 3.

Table 3

System parameters for simulation and load specifications.

Parameter	Value
Capacitance of the capacitor C_{dc}	5 mF
Coupling impedance $R_f L_f$	0.1 m Ω , 0.1 mH
The source voltage and frequency	220 V, 50 Hz
Source impedance $R_s L_s$	1 m Ω , 1 mH
Line impedance $R_l L_l$	1 m Ω , 1 mH
Load impedance $R_{ch} L_{ch}$	5 Ω , 10 mH
Unbalanced load $R_l L$	5 Ω , 10 mH

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